IB PHYSICS E.1 STRUCTURE OF THE ATOM

Geiger-Marsden-Rutherford experiment & discovery of the nucleus

- **Alpha particle scattering** → key evidence for nuclear structure.
- Thin gold foil bombarded with **α particles**; ZnS screen detected flashes.
- **Most particles passed straight** → atom mostly empty space.
- **Few deflected at large angles** → strong repulsion from small, dense region.
- **Some rebounded** → concentrated positive charge.
- Rutherford concluded **nucleus = small, massive, positively charged core**.



Note!

Thomson's **plum pudding model** predicted only small deflections → experiment contradicted it.

Nuclear notation

- Nuclei represented as $${}^{A}_{Z}X$$
  - **X** = chemical symbol
  - **Z** = proton (atomic) number
  - **A** = nucleon number (protons + neutrons)
- **Number of neutrons** = A − Z

Nuclear radius & density (HL)

- **Nuclear radius**: $$R = R_0 A^{\frac{1}{3}}$$
- $R_0 \approx 1.2 \times 10^{-15}$ m
- **A** = nucleon number
- **Volume** ∝ **A**, **mass** ∝ **A** ⇒ nuclear **density ≈ constant** for all nuclei.
- Typical density ≈ $2\text{–}3 \times 10^{17}$ kg m^{-3}.

High-energy scattering & distance of closest approach (HL)

- At high α energies → **deviations from Rutherford**:

$$\frac{1}{4\pi\epsilon_0}\frac{Q_\alpha Q_{\text{nucleus}}}{r_{\min}} = E_{\text{kinetic initial}}$$

- Inelastic scattering → nuclear excitation.
- Nuclear forces → deviations from pure Coulomb law.
- **Distance of closest approach** (head-on collision):
- Gives **upper limit** on nuclear radius.

Quantization of angular momentum (HL)

- **Bohr postulate**: $$mvr = \frac{nh}{2\pi}$$
- Only specific orbits allowed → discrete energies.
- Explains **stability** of orbits (no spiraling in).
- Underpins discrete spectral lines.

Emission & absorption spectra

- **Electrons** occupy discrete **energy levels**.
- **Emission spectrum** → bright lines produced when electrons **fall** to lower levels → emit photons.



- **Absorption spectrum** → dark lines where electrons **absorb** photons of exact energy to move up.



- **Spectral lines unique** to each element → "fingerprints" for identification.
- Photon energy: $$E = hf = \frac{hc}{\lambda}$$
- **h** = Planck's constant
- **f** = frequency
- **λ** = wavelength



Note!

Absorption lines occur at **same wavelengths** as emission lines for the given element.

Photon emission & absorption

- **Electron drops** from high to low → emits photon (E = hf).
- **Electron rises** to higher level → absorbs photon of matching energy.
- **Frequency** depends on energy difference between levels.
- **Spectral analysis** reveals **elemental composition** of stars, gases, etc.



Exam Tip!

- **Higher ΔE** → higher f, shorter λ (UV)
- **Lower ΔE** → lower f, longer λ (IR).

Bohr model energy levels (HL)

- **Energy levels in hydrogen**: $$E_n = -\frac{13.6}{n^2}\text{eV}$$
- **Negative sign** → bound energy; **n = 1** = ground state.
- **Transitions**: $$\Delta E = E_{\text{upper}} - E_{\text{lower}} = hf$$
- → determines spectral line frequency.

Scan the QR code for full IB Physics notes & practice questions on **revisiondojo.com**